## Exercises

**Exercise 10.1 · Vocabulary.** A graph $G$ has vertices 1, 2, 3, 4, 5, 6 and edges 1-2, 1-3, 2-3, 3-4, 4-5, 4-6, 5-6.

1. Give the order of $G$ and the degree of each vertex, and check the property of the sum of the degrees.
2. Is $G$ complete? connected?
3. Give a walk of length 3 from 1 to 6 and a cycle through 5.

**Solution.**

1. Order 6. Degrees: 2, 2, 3, 3, 2, 2, with sum $14 = 2 \times 7$.
2. Not complete (1 and 4 are not adjacent). Connected: every vertex is joined to 3 or to 4, and 3-4 is an edge.
3. 1-3-4-6; the cycle 4-5-6-4.

**Exercise 10.2 · Handshakes.**

1. Can one build a graph of order 7 all of whose vertices have degree 3?
2. Can six people each shake hands with exactly three others? If so, how many handshakes are there?

**Solution.**

1. No: the sum of the degrees would be 21, an odd number.
2. Yes, for example by placing the people at the vertices of a hexagon and joining each to its two neighbours and to the opposite person. The sum of the degrees is 18: there are 9 handshakes.

**Exercise 10.3 · A directed graph.** A directed graph has vertices $A$, $B$, $C$ and arcs $A \to B$, $A \to C$, $B \to C$ and $C \to A$.

1. Write its adjacency matrix $M$ (order $A$, $B$, $C$). Is it symmetric?
2. Compute $M^2$ and $M^3$.
3. How many paths of length 3 are there from $A$ to $A$? Write them out.

**Solution.**

1. The matrix is not symmetric:
$$M = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

2. We obtain
$$M^2 = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \qquad \text{and} \qquad M^3 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$
3. The entry $(A, A)$ of $M^3$ is 1: the path $A \to B \to C \to A$.

**Exercise 10.4 · A tournament.** Eight teams play a tournament: each meets each of the others once. Model with a graph and determine the number of matches.

**Solution.** Complete graph of order 8 (vertices: teams; edges: matches). There are $\binom{8}{2} = 28$ matches.

**Exercise 10.5 · Degrees and the square of the matrix.** A graph has adjacency matrix
$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$
1. Draw it.
2. Compute $M^2$ and compare its diagonal with the degrees of the vertices.
3. Prove that for every undirected graph, $\left(M^2\right)_{ii}$ is the degree of vertex $i$.

**Solution.**

1. Edges: 1-2, 1-3, 2-3, 2-4, 3-4.
2. The diagonal of $M^2$, namely 2, 3, 3, 2, gives the degrees:
$$M^2 = \begin{pmatrix} 2 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 1 & 2 & 3 & 1 \\ 2 & 1 & 1 & 2 \end{pmatrix}.$$
3. $\left(M^2\right)_{ii} = \sum_k m_{ik} m_{ki} = \sum_k m_{ik}^2 = \sum_k m_{ik}$, since $m_{ki} = m_{ik}$ and $m_{ik} \in \{0, 1\}$. This is the number of neighbours of $i$. Another reading: a closed walk of length 2 starting from $i$ is a there-and-back trip along an edge leaving $i$.

**Exercise 10.6 · Testing connectedness.** A graph has vertices 1 to 5 and edges 1-2, 2-3, 3-1, 4-5. Let $M$ be its adjacency matrix.

1. Compute, with a calculator, $S = M + M^2 + M^3 + M^4$.
2. Is the graph connected? Justify using $S$, then directly.

**Solution.**

1. We obtain
$$S = \begin{pmatrix} 10 & 10 & 10 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}.$$
2. No: the entry $(1,4)$ is zero, so no walk of length at most 4 joins 1 to 4, and by the course no walk at all. Directly: the vertices 1, 2, 3 are only joined to one another.

**Exercise 10.7 · Counting triangles.** A triangle of a graph is a set of three pairwise adjacent vertices.

1. Prove that the number of triangles is $\frac{1}{6}$ of the sum of the diagonal entries of $M^3$.
2. Apply this to the complete graph of order 4 and to the graph of exercise 10.5.

**Solution.**

1. $\left(M^3\right)_{ii}$ counts the closed walks of length 3 starting from $i$. Such a walk $i$-$j$-$k$-$i$ has three distinct vertices (there is no edge from a vertex to itself), which form a triangle. Each triangle gives 6 closed walks of length 3: 3 choices of the starting vertex and 2 directions.
2. For the complete graph of order 4, the diagonal of $M^3$ is 6, 6, 6, 6: $\frac{24}{6} = 4$ triangles, that is $\binom{4}{3}$. For the graph of exercise 10.5: 2 triangles, $\{1,2,3\}$ and $\{2,3,4\}$.

## Problem

**Problem · Drawing without lifting the pencil.** An *Eulerian trail* is a walk that uses every edge of the graph exactly once.

1. Assume that a graph has an Eulerian trail whose ends are distinct. Prove that its two ends have odd degree and that all the other vertices have even degree. What can be said if the trail is closed?
2. We take the converse for granted (Euler's theorem): a connected graph all of whose vertices have even degree, except zero or two, has an Eulerian trail; if there are two vertices of odd degree, it starts at one and ends at the other.

   Consider the "house": a square $abdc$ ($a$ and $b$ at the bottom, $c$ and $d$ at the top) with its two diagonals, topped by a roof $c$-$e$-$d$. List the edges and the degrees.
3. Can the house be drawn without lifting the pencil or going over a line twice? If so, where must one start? Give such a trail.
4. Can it be done returning to the starting point?

**Solution · Problem.**

1. Each passage through an intermediate vertex uses one edge to arrive and another to leave: it contributes 2 to the degree. Since every edge is used once, a vertex that is not an end has even degree. An end has in addition one unpaired edge of departure (or arrival): its degree is odd. If the trail is closed, all the degrees are even.
2. Edges: $a$-$b$, $a$-$c$, $b$-$d$, $c$-$d$, $a$-$d$, $b$-$c$, $c$-$e$, $d$-$e$ (eight edges). Degrees: $a$: 3, $b$: 3, $c$: 4, $d$: 4, $e$: 2.
3. The graph is connected with exactly two vertices of odd degree, $a$ and $b$: yes, starting from $a$ (or from $b$). For example $a$-$b$-$c$-$a$-$d$-$c$-$e$-$d$-$b$, which uses each of the eight edges once.
4. No: all the degrees would have to be even.

**Teacher note · Status of the problem.** Eulerian graphs are a "possible problem" of the syllabus. Euler's theorem is taken for granted in its sufficient part. Extension: the problem of the bridges of Königsberg; the site's workshop on drawing without lifting the pencil covers the same theme.